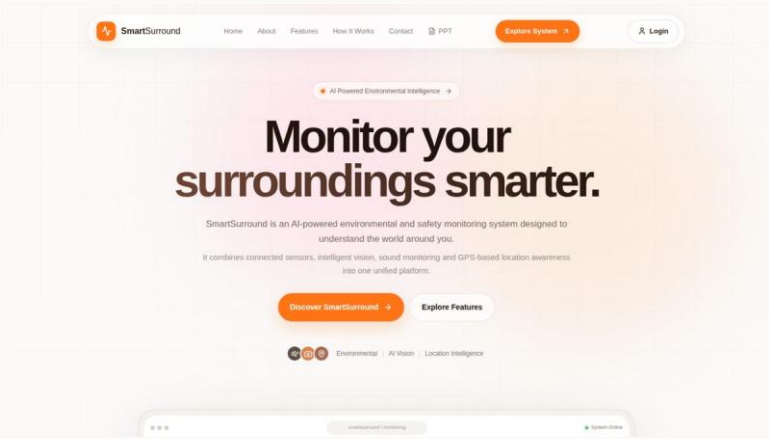


SmartSurround

AI-Enabled IoT-Based Real-Time Environmental and Surrounding Safety Monitoring System

01 INPUT	Sensors · Camera · GPS
02 INTELLIGENCE	Cloud AI · Thresholds
03 OUTPUT	Dashboard · Alerts · Reports



The live SmartSurround landing page

TEAM ABISKAR

Software Engineering Laboratory

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Roll:- 66
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Problem Statement & Research Gap

Why monitoring must move from reactive checks to continuous, connected safety intelligence

THE PROBLEM

Environmental monitoring is reactive: manual checks, delayed detection, and warnings that arrive only after an unsafe condition has occurred.

THE GAP

Sensor readings, camera footage and location usually live in separate tools, with no single real-time view tying a risk to where it happens.

THE RESPONSE

SmartSurround fuses ESP32 sensors, a live camera, GPS and cloud AI into one dashboard with proactive threshold alerts.

Existing pattern	Limitation	SmartSurround approach
Manual periodic checks	Delayed detection	Continuous sensing with a live dashboard
Isolated sensors	No location context	GPS-tagged readings and events
Raw numbers only	User must interpret values	Status labels + editable alert thresholds
Warnings after the event	Reactive, not preventive	Proactive threshold alerts
Unverified road complaints	Noise and false reports	AI detection + corroboration + admin verification

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How can live sensor data, camera vision and GPS context be fused into a secure, explainable, real-time safety monitoring and alerting platform?

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Eight core capabilities of the SmartSurround website, mapped to functional requirements

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Secure Access

Firestore email sign-up with verification, plus a separate admin login.

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Live Air & Environment

IAQ, PM1.0 / 2.5 / 10, CO₂, temperature and humidity from the ESP32.

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MJPEG stream from the ESP32-CAM, auto-discovered via heartbeat.

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GPS Location

Latitude, longitude, altitude, speed, satellites, HDOP and fix type.

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Threshold Alerts

Editable limits turn live readings into warning and danger alerts.

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Start / stop recording at a chosen interval and export to Excel.

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AI Road Damage

YOLOv8 finds potholes and cracks; reports are corroborated and routed to authorities.

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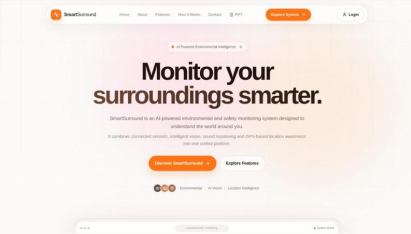
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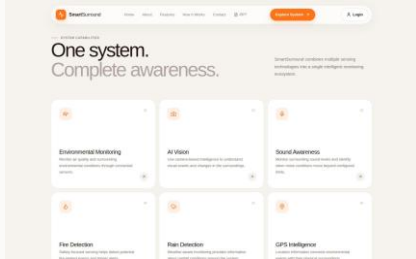
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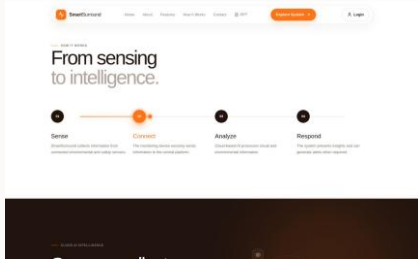
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Technical, operational, economic, security and schedule feasibility — Agile delivery

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React + Vite UI, Flask backend, ESP32 firmware, Firebase and a YOLOv8 model.

Feasible

Operational

Sense → Connect → Analyze → Respond; one dashboard for users and one control center for admins.

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Low-cost ESP32 boards and sensors; hosting on Vercel / GitHub Pages and Render.*

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Email-verified accounts, admin PIN with token and lockout; device endpoints still need hardening.

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GPS shared only after device permission.

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No fake data: fields show "--" until real values arrive.

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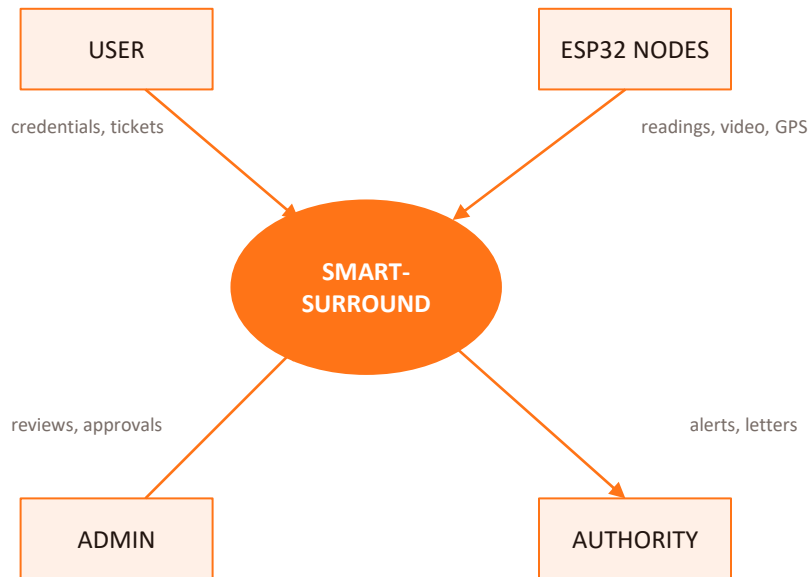
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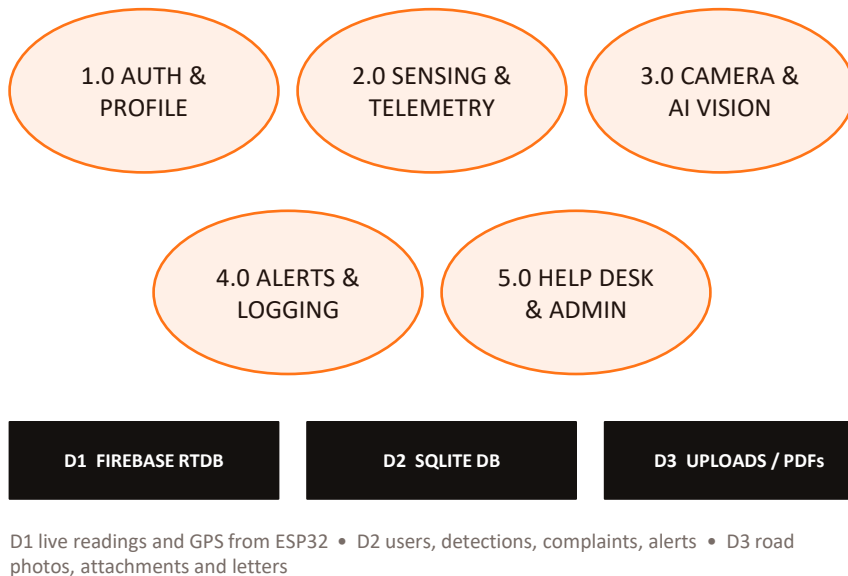
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ACTION	Queue: AI confident	Queue: AI uncertain	Queue: unclassified (Warning)	Normal road: not surfaced

Sensor alert (defaults)	Warning	Danger
PM2.5	35	55
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Noisy-OR across every raw box: several weak signals agreeing still counts. Severity: Potholes and Alligator Crack = Critical; Longitudinal and Transverse Crack = Warning.

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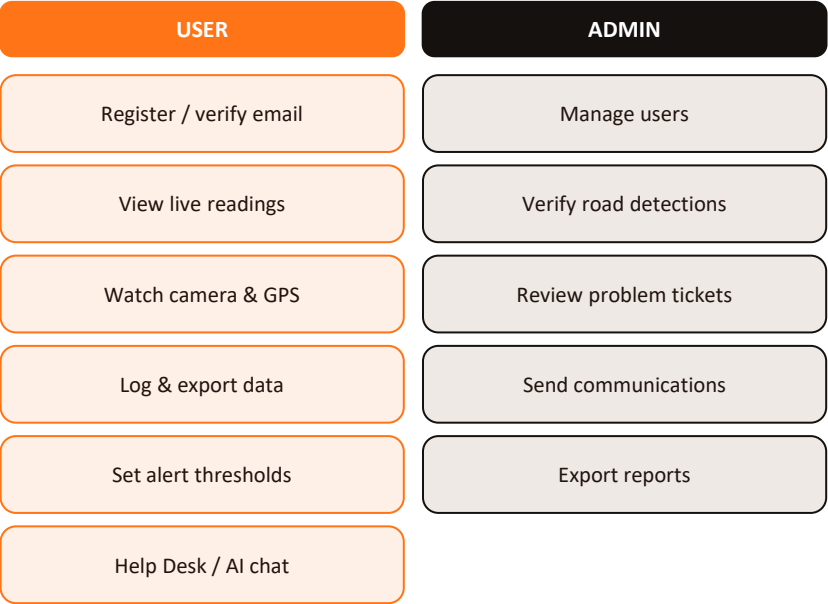
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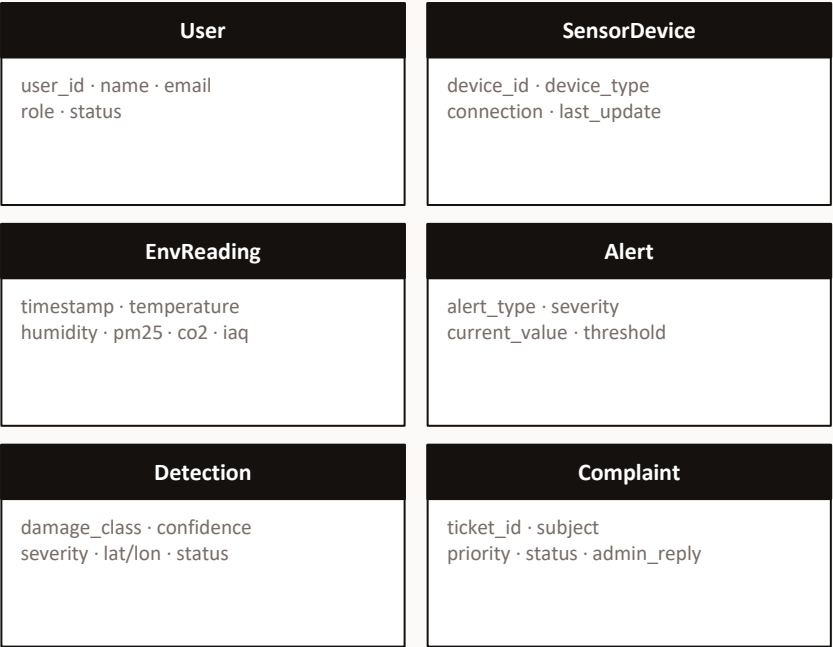
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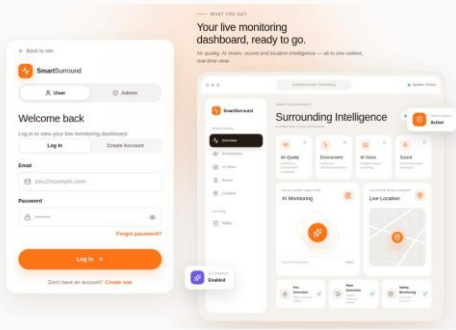
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The real login screen: User / Admin tabs, Log In and Create Account

ADMIN: PIN login → token → control center; every admin action needs the token.

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Only email-verified Firebase users reach /dashboard; unverified or signed-out visitors are redirected to /login.

Website Demo — Dashboard & Admin Control Center

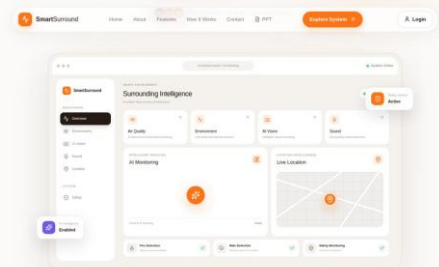
What each page shows after login (every value comes from the connected ESP32; no simulated data)

USER DASHBOARD

- Overview — IAQ gauge, PM, temp, humidity, CO2
- Air Quality — Trend chart, adjustable range
- Environment — Detail and comfort score
- Camera — Live MJPEG from ESP32-CAM
- Location — GPS fix from NEO-8M
- Data Log — Record and export to Excel
- Alerts — Live alerts, editable limits
- Help Desk · Profile · Settings — Tickets, avatar, theme

ADMIN CONTROL CENTER

- Dashboard
- Live Operations: map, users, history
- Users: database, details, activity
- Monitoring: air, cameras, sensors, alerts
- Problem Desk: inbox, verification, history
- Reports: environment, GPS, Excel export
- Communications: AI drafts, sent history
- System: thresholds, activity, settings



Dashboard preview on the landing page

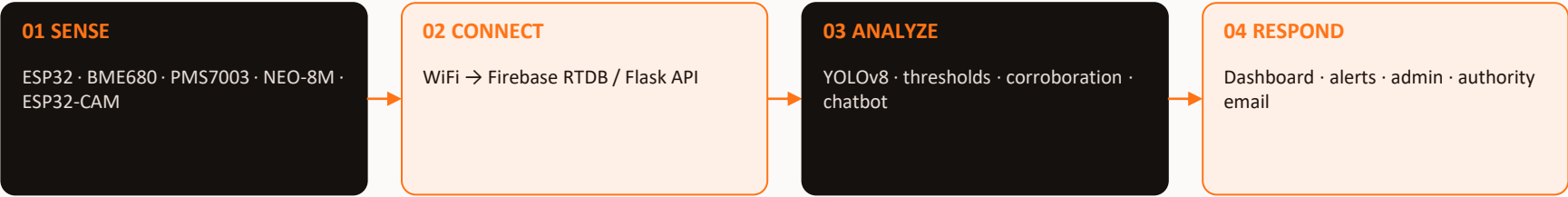
LIVE DEMO TIP

Log in, open Overview and Camera with both ESP32 boards powered on. Then open Admin to show the same data from the control center.

Technology Stack & System Architecture

Layer-by-layer technology choices and the four-layer implementation architecture

Layer	Technology	Purpose
UI	React 19 + Vite, Tailwind 4, Framer Motion	Landing site, dashboard and admin pages
Backend	Python Flask 3 + gunicorn (Render)	APIs, admin auth, detection workflow
Identity	Firebase Authentication	Email / password with verification
Live data	Firebase Realtime Database	ESP32 readings and GPS to dashboard
Database	SQLite + openpyxl	Users, alerts, tickets, detections, Excel reports
AI	YOLOv8 (Ultralytics) + hosted chatbot	Road-damage detection and Q&A assistant
Hardware	ESP32, ESP32-CAM, BME680, PMS7003, NEO-8M	Sensing, video and location
Messaging	Resend email + reportlab PDFs	Authority email and letters



Implementation — Service Modules & Security Design

Separated, maintainable services and six built-in access controls

SERVICE MODULES

1

Auth Service

Firebase Auth for users; admin PIN with a 30-minute httpOnly token.

2

Telemetry Service

ESP32 reads BME680, PMS7003 and NEO-8M; pushes to Firebase and local API.

3

Vision & AI Service

detector.py runs YOLOv8; camera frames analysed via `/api/camera/analyze`.

4

Reporting Service

Corroboration, authority routing, email driver and PDF letters.

5

Support Service

Help Desk tickets, notices, AI chatbot and the admin control center.

SECURITY CONTROLS

✓ Email-verified access

Dashboard routes open only to verified Firebase accounts.

✓ Verified ID tokens

Backend checks the Firebase token before storing user GPS.

✓ Admin PIN gate

Salted SHA-256 PIN; admin routes need `require_auth`.

✓ Lockout

5 failed PINs from one IP block it for 1 hour.

✓ Exact-origin CORS

Backend allows only listed frontend origins.

✓ Upload tokens

`/upload` needs a token from `/upload/token`.

Review point: ESP32 endpoints have no authentication and WiFi / database credentials sit in firmware source — move them to API keys and secret storage before production.

SDG Alignment, Social Impact & Oversight

Sustainable Development Goals, implementation checklist and safeguards

SDG 3

GOOD HEALTH & WELL-BEING

Air-quality monitoring (PM, CO2, IAQ) flags unhealthy conditions early.

SDG 9

INDUSTRY, INNOVATION & INFRASTRUCTURE

Low-cost IoT sensors plus cloud AI as monitoring infrastructure.

SDG 11

SUSTAINABLE CITIES & COMMUNITIES

Road-damage detection and authority reporting improve public safety.

Area	Checklist item	Status
Security	Verified accounts, admin token, lockout, exact-origin CORS	✓
Privacy	GPS stored only after device permission; no simulated data	✓
Social impact	Cuts delay between a hazard and a response	✓
Accessibility	Responsive layout, light / dark / system theme, language setting	✓
Risk	Road model is research-grade (mAP@0.5 $\approx$ 0.55); needs local fine-tuning	Attention

Transparency

AI confident / uncertain badges

Human review

Admin approve or reject

Support

Help Desk + chatbot

Admin oversight

Users, activity, tickets

CONCLUSION

SmartSurround — Sense, Understand, Respond

- Continuous sensing replaces manual checks, so risks are flagged before they escalate.
- Sensors, camera, GPS and cloud AI meet in one live dashboard and one admin control center.
- AI detections stay explainable and reviewable: confidence badges, corroboration and admin approval.
- Next: cloud relay for live data anywhere, edge AI, multi-post control and a mobile app.

FINAL SUMMARY

PROBLEM

Reactive monitoring

APPROACH

Sensors + AI + GPS

STACK

React · Flask · ESP32

SDGS

3 · 9 · 11

SDLC

Agile

Goal: intelligent sensing, detection, prediction and safety for every monitored surrounding.

THANK YOU FOR YOUR TIME

Thank You

Safer surroundings begin with smarter awareness.

Technology matters most when it protects people. We built SmartSurround to turn data into timely, trusted action.

PRESENTED BY

TEAM ABISKAR

Software Engineering Laboratory

01 SENSE

Sensors, camera and GPS watch every surrounding.

02 UNDERSTAND

Cloud AI and thresholds turn readings into insight.

03 RESPOND

Dashboards, alerts and reports drive timely action.

QUESTIONS & FEEDBACK

We would be glad to hear your thoughts.

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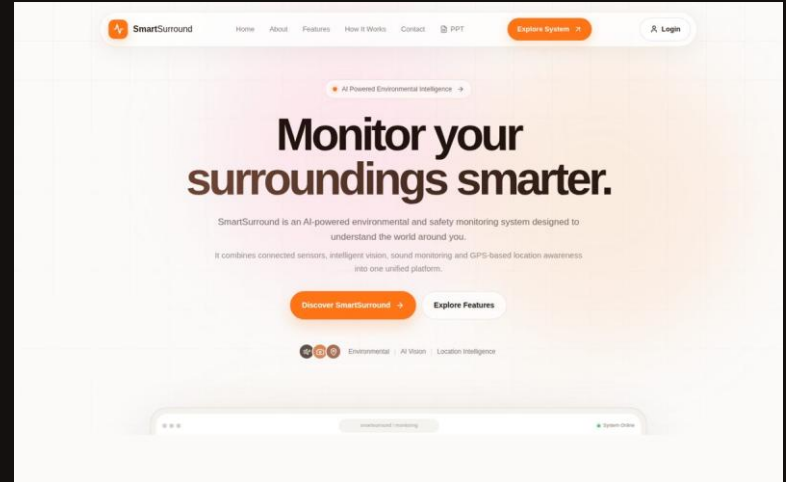
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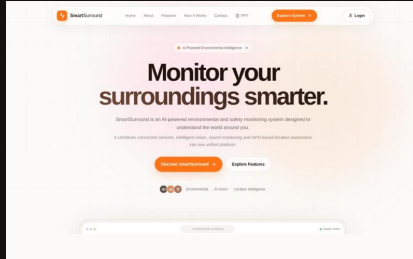
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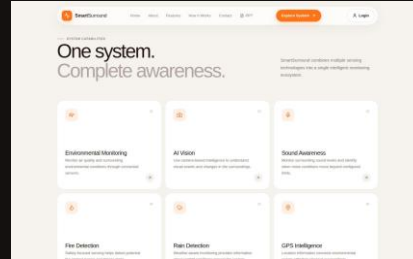
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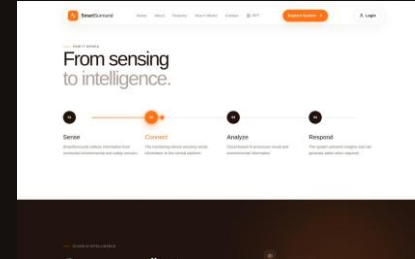
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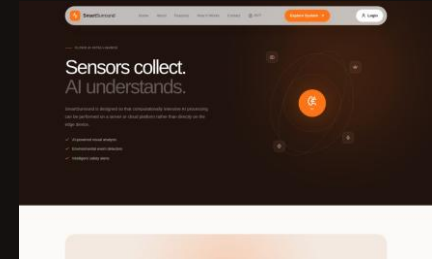
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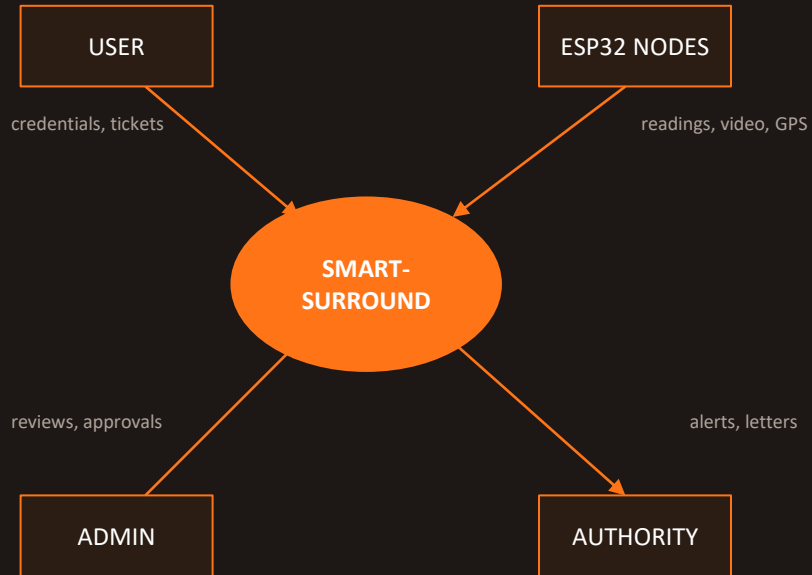
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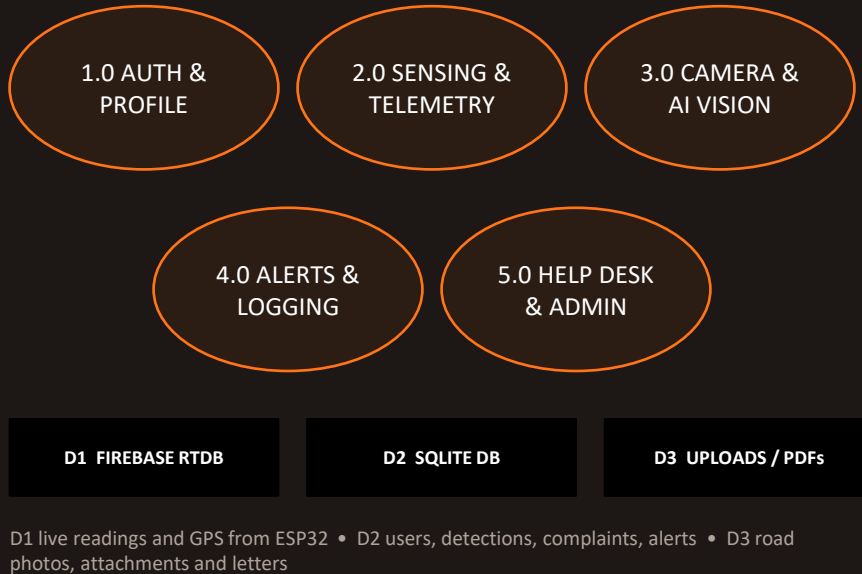
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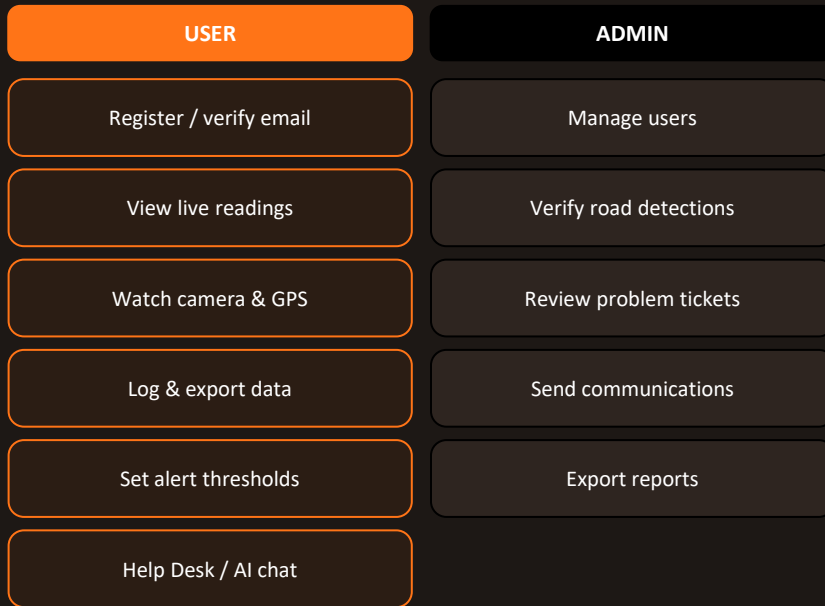
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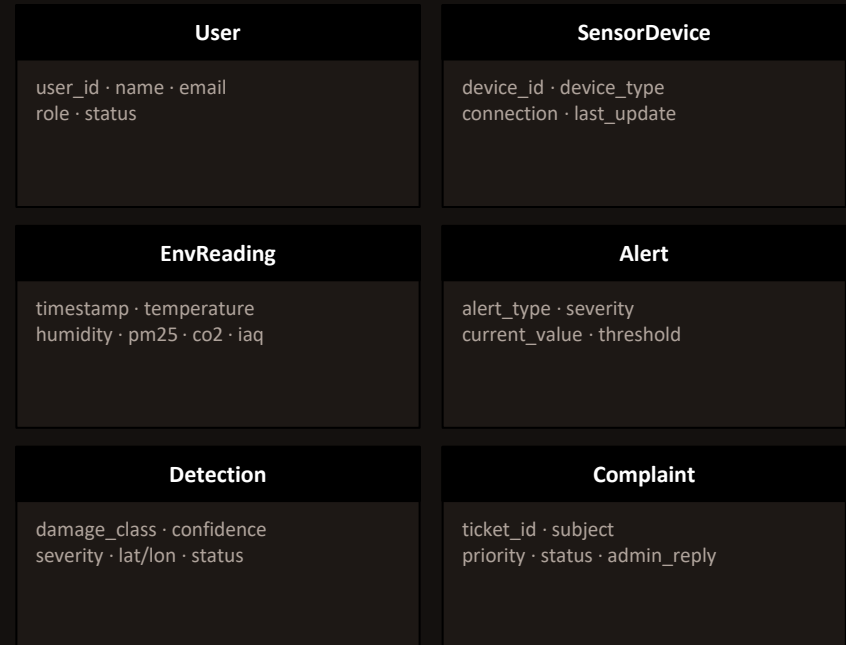
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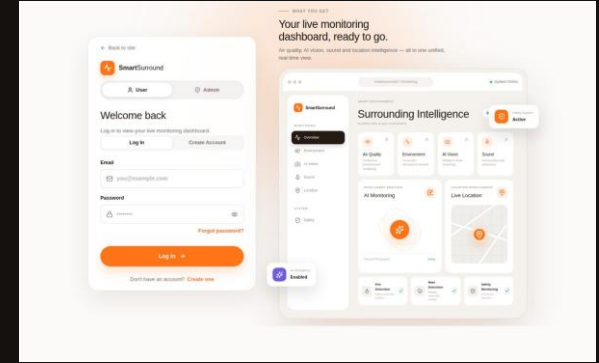
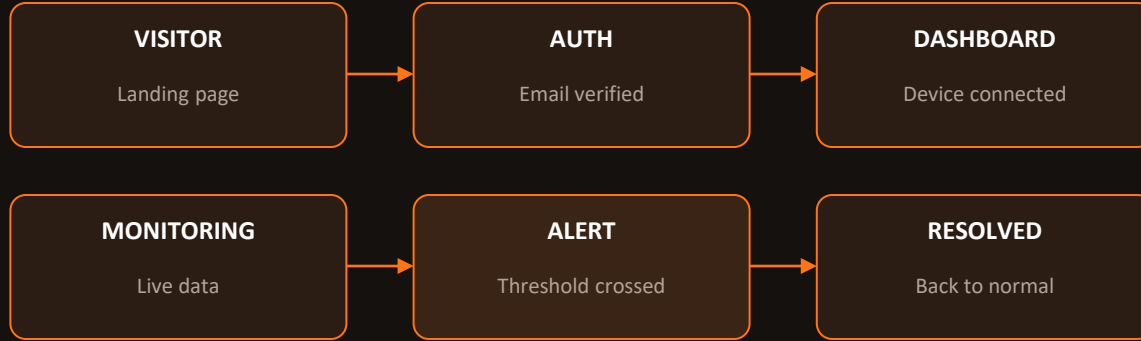
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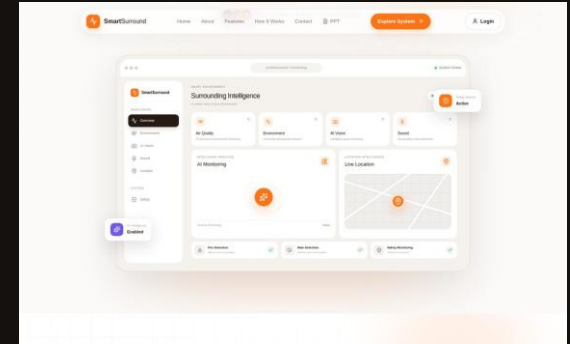
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AI	YOLOv8 (Ultralytics) + hosted chatbot	Road-damage detection and Q&A assistant
Hardware	ESP32, ESP32-CAM, BME680, PMS7003, NEO-8M	Sensing, video and location
Messaging	Resend email + reportlab PDFs	Authority email and letters

01 SENSE

ESP32 · BME680 · PMS7003 · NEO-8M ·
ESP32-CAM

02 CONNECT

WiFi → Firebase RTDB / Flask API

03 ANALYZE

YOLOv8 · thresholds · corroboration ·
chatbot

04 RESPOND

Dashboard · alerts · admin · authority
email

Implementation — Service Modules & Security Design

Separated, maintainable services and six built-in access controls

SERVICE MODULES

1

Auth Service

Firebase Auth for users; admin PIN with a 30-minute httpOnly token.

2

Telemetry Service

ESP32 reads BME680, PMS7003 and NEO-8M; pushes to Firebase and local API.

3

Vision & AI Service

detector.py runs YOLOv8; camera frames analysed via `/api/camera/analyze`.

4

Reporting Service

Corroboration, authority routing, email driver and PDF letters.

5

Support Service

Help Desk tickets, notices, AI chatbot and the admin control center.

SECURITY CONTROLS

✓ Email-verified access

Dashboard routes open only to verified Firebase accounts.

✓ Verified ID tokens

Backend checks the Firebase token before storing user GPS.

✓ Admin PIN gate

Salted SHA-256 PIN; admin routes need `require_auth`.

✓ Lockout

5 failed PINs from one IP block it for 1 hour.

✓ Exact-origin CORS

Backend allows only listed frontend origins.

✓ Upload tokens

`/upload` needs a token from `/upload/token`.

Review point: ESP32 endpoints have no authentication and WiFi / database credentials sit in firmware source — move them to API keys and secret storage before production.

SDG Alignment, Social Impact & Oversight

Sustainable Development Goals, implementation checklist and safeguards

SDG 3

GOOD HEALTH & WELL-BEING

Air-quality monitoring (PM, CO2, IAQ) flags unhealthy conditions early.

SDG 9

INDUSTRY, INNOVATION & INFRASTRUCTURE

Low-cost IoT sensors plus cloud AI as monitoring infrastructure.

SDG 11

SUSTAINABLE CITIES & COMMUNITIES

Road-damage detection and authority reporting improve public safety.

Area	Checklist item	Status
Security	Verified accounts, admin token, lockout, exact-origin CORS	✓
Privacy	GPS stored only after device permission; no simulated data	✓
Social impact	Cuts delay between a hazard and a response	✓
Accessibility	Responsive layout, light / dark / system theme, language setting	✓
Risk	Road model is research-grade (mAP@0.5 $\approx$ 0.55); needs local fine-tuning	Attention

Transparency

AI confident / uncertain badges

Human review

Admin approve or reject

Support

Help Desk + chatbot

Admin oversight

Users, activity, tickets

CONCLUSION

SmartSurround — Sense, Understand, Respond

- Continuous sensing replaces manual checks, so risks are flagged before they escalate.
- Sensors, camera, GPS and cloud AI meet in one live dashboard and one admin control center.
- AI detections stay explainable and reviewable: confidence badges, corroboration and admin approval.
- Next: cloud relay for live data anywhere, edge AI, multi-post control and a mobile app.

FINAL SUMMARY

PROBLEM

Reactive monitoring

APPROACH

Sensors + AI + GPS

STACK

React · Flask · ESP32

SDGS

3 · 9 · 11

SDLC

Agile

Goal: intelligent sensing, detection, prediction and safety for every monitored surrounding.

THANK YOU FOR YOUR TIME

Thank You

Safer surroundings begin with smarter awareness.

Technology matters most when it protects people. We built SmartSurround to turn data into timely, trusted action.

PRESENTED BY

TEAM ABISKAR

Software Engineering Laboratory

01 SENSE

Sensors, camera and GPS watch every surrounding.

02 UNDERSTAND

Cloud AI and thresholds turn readings into insight.

03 RESPOND

Dashboards, alerts and reports drive timely action.

QUESTIONS & FEEDBACK

We would be glad to hear your thoughts.